

ADITYA CHOUHAN

Java Backend Developer | Software Engineer

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PROFESSIONAL SUMMARY

Results-driven **Java Developer** (Java 21, Spring Boot 3.3) with expertise in engineering scalable backend architectures, distributed microservices, and high-throughput RESTful APIs. Solid foundation in **Data Structures & Algorithms (DSA)**, **Object-Oriented Programming (OOPs)**, and relational database design (PostgreSQL, MySQL 3NF/ACID, Redis). Hands-on experience integrating **Spring AI**, event-driven messaging with **Apache Kafka**, and containerizing services with **Docker** for cloud deployment.

TECHNICAL SKILLS

Programming Languages: Java (Java 21 / 17), JavaScript (ES6+), C++, SQL, HTML5, CSS3
Frameworks & Backend: Spring Boot 3.3, Spring Security, Spring AI, Hibernate / JPA, RESTful APIs, Microservices, JDBC, JSP, Java Collection API
Databases & Caching: PostgreSQL, MySQL 8.0 (3NF/ACID), MongoDB, Redis (In-Memory / Cache / Queue), Supabase
DevOps, Cloud & Tools: Docker, Apache Kafka, Git, GitHub, Maven, Gradle, Postman, JUnit 5, Vercel, Render, IntelliJ IDEA, VS Code
Core Competencies: Data Structures & Algorithms (DSA), OOPs, System Design Basics, Database Indexing & Optimization, JWT & OAuth2, CI/CD Basics

KEY TECHNICAL PROJECTS

FairCart — AI-Powered Distributed E-Commerce & Smart Cart Platform

Java 21, Spring Boot 3.3, Spring AI, Microservices, Apache Kafka, Redis, PostgreSQL, Hibernate, Docker, JWT

- **Distributed Microservices & Messaging:** Architected decoupled microservices (Auth, Catalog, Cart, Order, Inventory) communicating asynchronously via **Apache Kafka** event streams with Dead-Letter Queues (DLQ) for fault-tolerant order dispatching.
- **Spring AI & Dynamic Deals:** Integrated **Spring AI** with LLM models for real-time dynamic deal negotiation, smart product recommendations, and automated cart abandonment anomaly detection.
- **High-Throughput Caching Layer:** Built a distributed session and cart caching mechanism using **Redis**, cutting primary database read load by **55%** and achieving sub-**25ms** response latency under concurrent flash-sale traffic.
- **ACID Transactions & Data Integrity:** Designed normalized (3NF) relational schemas in **PostgreSQL** using Hibernate/JPA, implementing pessimistic locking to eliminate race conditions during high-concurrency inventory checkouts.
- **Stateless Security & Containerization:** Enforced stateless **JWT & OAuth2** authentication with granular Role-Based Access Control (RBAC); automated unit testing with **JUnit 5 & Mockito** and containerized services using **Docker**.

FixMate — On-Demand Local Service Booking & Management Platform

Java 21, Spring Boot 3.3, Spring Security, JWT, MySQL 8.0, Redis, Tailwind CSS, Docker, Render

- **Multi-Role Marketplace Architecture:** Built a production-grade service marketplace featuring dedicated dashboards (Customer, Service Pro, Admin) secured by stateless **JWT** authentication and **BCrypt** password encryption.
- **Smart Geo-Proximity & Slot Booking:** Implemented location-aware service discovery using the **Haversine formula** across target city clusters (Ujjain, Indore, Bhopal) paired with dynamic, collision-free slot reservation logic.
- **High-Performance Relational Schema:** Architected **17 normalized (3NF)** tables in **MySQL 8.0** with composite B-Tree indexing, foreign key cascading, and ACID stored procedures for dispute handling and transaction auditing.
- **Job Lifecycle & Cloud Deployment:** Engineered an automated 5-stage job lifecycle pipeline (BOOKED → ASSIGNED → IN_PROGRESS → COMPLETED → PAID) with real-time status tracking; containerized with **Docker** and deployed on **Render**.

EDUCATION

Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore, MP Bachelor of Technology (B.Tech) in Computer Science & Engineering	2022 – 2026 CGPA: 6.98 / 10.0
Ujjain Public School, Ujjain, MP Class XII (Senior Secondary) – Science (PCM) MP Board	2021 – 2022 Higher Secondary
Jolly Memorial Mission School, Ujjain, MP Class X (Secondary School) CBSE Board	2019 – 2020 Secondary School

CERTIFICATIONS & INTERESTS

Certifications: Java Programming (Specialization) • Cloud Computing – NPTEL / IIT Kharagpur • Relational Databases & MySQL
Interests: Competitive Programming, Data Structures & Algorithms, Exploring Generative AI & Cloud Systems, Open Source Contributions